WATERWATCH · USER REQUIREMENTS

WaterWatch user requirements

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DOCUMENT CONTROL

Document ID	WWT-URS-001	Version	0.1
Status	Draft	Date	2026-10-25
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Project TRL	TRL 3 Proof of concept		
Repository	github.com/BoujeeEnjinia1701/waterwatch		
Source	docs/03a-user-requirements.md @ 59e5426		

REVISION HISTORY

Version	Date	Author	Change
0.1	2026-10-25	Amish Chadha	First user requirements, drawn from WWT-PRB-001 Tables 1 to 3 (kit 1.9.1 upgrade)

WaterWatch user requirements

What caretakers, scheme operators, district officers, researchers and the people who drink the water need, written from their side. They are drawn from the users and stakeholders in WWT-PRB-001 (Tables 1 to 3) and the decisions Amish has made (WWT-DDR-001, WWT-DDR-002, WWT-DEC-001). None is yet validated with a scheme operator; co-design with the first partner will revise them. The product requirements that meet these needs are in WWT-PRS-001 (`03-requirements.md`), and the links are in the traceability matrix WWT-RTM-001 (`03b-traceability.md`). Method: STANDARDS section 22.

Table 1. User requirements.

ID	Category	Title	Need	Priority	Source
UR-001	Performance	Chlorine at the tap	Know whether free chlorine is reaching the tap, often enough to act before people drink unprotected water for days	High	Caretaker; scheme operator; district health officer (PRB Tables 1 and 2)
UR-002	Performance	Cloudy water	Know when the water turns cloudy, because chlorine stops working well in turbid water	High	Scheme operator; district water officer
UR-003	Usability	Alerts to the people who act	Get a plain alert on a basic phone soon after a problem is confirmed, in the local language	High	Caretaker; scheme operator (WWT-DEC-001, alert rules)
UR-004	Performance	Trends and records	See readings from many tapstands over time, kept even when the network fails, in an open format	Medium	Scheme operator; district officers
UR-005	Environment and durability	No mains power	Run on sunlight alone, through cloudy weeks, with no mains supply at the site	High	Scheme operator (PRB, Users and context)
UR-006	Environment and durability	Outdoor site	Keep working in direct sun, heat to 45 °C, dust and splashing for years	High	Scheme operator; caretaker
UR-007	Safety	Supply kept clean	Never contaminate the drinking water supply, by backflow or by unsafe materials	High	People who drink the water; district health officer (PRB Table 3)
UR-008	Logistics	Little water used	Use only a small share of the treated water the tapstand delivers	Medium	Water committee (PRB, Constraints)
UR-009	Maintenance	Monthly caretaker check	Keep the unit accurate with one short monthly visit by a trained caretaker using a comparator kit and no special tools	High	Caretaker; scheme operator
UR-010	Deployment and assembly	Simple installation	Install beside an existing tapstand with hand tools, changing nothing on the tapstand beyond one tee on the riser	High	Scheme operator; water committee (PRB, Constraints)
UR-011	Cost	Affordable per point	Cost little enough per tapstand that a small operator can monitor many points	High	Scheme operator; funders (`project.yaml` value-engineering target)
UR-012	Regulatory	Measurements, not verdicts	Report what was measured and when, and never claim that the water is safe	High	District health officer (PRB, Constraints)
UR-013	Performance	Water temperature	Know the water temperature, for chlorine correction and context	Low	Researcher; scheme operator
UR-014	Usability			Medium	

ID	Category	Title	Need	Priority	Source
		Open and reproducible	Build, inspect and improve the design from published files		Researcher or open hardware community (PRB Table 2)
UR-015	Safety	Safe to handle	Handle the unit, its battery and its calibration chemicals without harm	High	Caretaker; technician
UR-016	Performance	Adjustable schedule	Change how often the unit reads, for an outbreak or a study	Low	Scheme operator; researcher

Categories: deployment and assembly, performance, usability, safety, environment and durability, logistics, maintenance, cost, regulatory. Priorities: High, Medium, Low.

> **Safety:** A unit that meets these needs still does not show that water is safe to drink. Accredited laboratory testing, including microbiological testing, remains necessary (WWT-PRC-001, Safety).